Cox Models

Model 1: Death

```
Call:
coxph(formula = Surv(TIME_30, EVENT_30 == 1) ~ ESCALATED, data = .)

  n= 4347, number of events= 187

                      coef exp(coef) se(coef)      z Pr(>|z|)
ESCALATEDEscalated -0.4724    0.6235   0.2270 -2.081   0.0374 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                   exp(coef) exp(-coef) lower .95 upper .95
ESCALATEDEscalated    0.6235      1.604    0.3996    0.9728

Concordance= 0.534  (se = 0.012 )
Likelihood ratio test= 4.88  on 1 df,   p=0.03
Wald test            = 4.33  on 1 df,   p=0.04
Score (logrank) test = 4.41  on 1 df,   p=0.04
```

Interpretation:

Escalation was significantly associated with the hazard of death/deterioration within 30 days after landmark time, using stable patients as the reference group (p-value = 0.037, z = -2.081). Escalated patients had an estimated 37.7% lower hazard of death/deterioration compared to stable patients.

Model 2: Transplantation

```
Call:
coxph(formula = Surv(TIME_30, EVENT_30 == 2) ~ ESCALATED, data = heart_data)

  n= 4347, number of events= 3093

                      coef exp(coef) se(coef)     z Pr(>|z|)
ESCALATEDEscalated 0.11052   1.11686  0.04551 2.429   0.0152 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                   exp(coef) exp(-coef) lower .95 upper .95
ESCALATEDEscalated     1.117     0.8954     1.022     1.221
```

```
Concordance= 0.51  (se = 0.004 )
Likelihood ratio test= 5.77  on 1 df,   p=0.02
Wald test            = 5.9  on 1 df,   p=0.02
Score (logrank) test = 5.9  on 1 df,   p=0.02
```

Interpretation:
Escalation was significantly associated with the hazard of transplantation within 30 days after landmark time, using stable patients as the reference group (p-value = 0.0152, z = 2.429). Escalated patients had an estimated 11.7% higher hazard of transplantation compared to stable patients.

Fine-Gray Subdistribution Models

Model 1: Death

```
Competing Risks Regression

Call:
crr(ftime = heart_matrix$TIME_30, fstatus = heart_matrix$EVENT_30,
    cov1 = covariance_matrix[, 1], failcode = 1)

                        coef exp(coef) se(coef)     z p-value
covariance_matrix[, 1]1 -0.535     0.586    0.227 -2.36   0.018

                      exp(coef) exp(-coef)  2.5% 97.5%
covariance_matrix[, 1]1     0.586       1.71 0.376 0.913

Num. cases = 4345
Pseudo Log-likelihood = -1550
Pseudo likelihood ratio test = 6.35  on 1 df,
```

Interpretation:
Escalated patients had a significantly lower subdistribution hazard of death/deterioration compared to stable patients, accounting for transplantation as a competing risk (p-value = 0.018, z = -2.36). Escalated patients had a subdistribution hazard of death/deterioration of 41.4% lower than stable patients (sHR = 0.586).

Model 2: Transplantation

```
Competing Risks Regression

Call:
crr(ftime = heart_matrix$TIME_30, fstatus = heart_matrix$EVENT_30,
```

```
    cov1 = covariance_matrix[, 1], failcode = 2)

                         coef exp(coef) se(coef)    z p-value
covariance_matrix[, 1]1 0.122      1.13   0.0446 2.73  0.0063

                        exp(coef) exp(-coef) 2.5% 97.5%
covariance_matrix[, 1]1      1.13      0.885 1.03  1.23

Num. cases = 4345
Pseudo Log-likelihood = -24417
Pseudo likelihood ratio test = 6.97  on 1 df,
```

Interpretation:
Escalated patients had a significantly higher subdistribution hazard of transplantation compared with stable patients, accounting for death/deterioration as a competing risk (p-value = 0.0063, z = 2.73). Escalated patients had a 13% higher subdistribution hazard of transplantation than stable patients (sHR = 1.13).